A Minor Project Synopsis

On

T211

Online ML-Based Academic Performance Analyzer

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DEGREE

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Title of the Project

Online ML-Based Academic Performance Analyzer

Introduction

Machine Learning has become an important tool in the education sector for analyzing and predicting student performance. Traditional evaluation methods only assess students after exams, which delays intervention. This project aims to develop an intelligent system that uses machine learning algorithms to analyze academic data and predict student performance in advance.

Problem Statement

Educational institutions often rely on traditional evaluation methods that assess student performance only after examinations. This delay prevents early identification of struggling students. Additionally, large volumes of student data generated through online and offline platforms remain underutilized.

There is a need for a system that can analyze this data efficiently, predict student performance in advance, and assist educators in making data-driven decisions.

Objectives

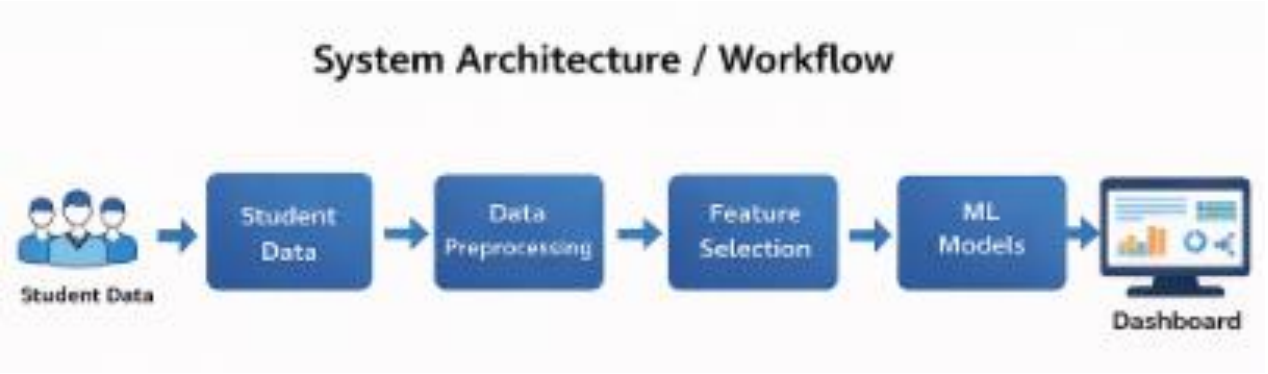
- To develop a machine learning-based system for predicting student performance
- To identify at-risk students at an early stage
- To improve prediction accuracy using feature engineering
- To compare multiple machine learning algorithms
- To provide visual insights through graphs and dashboards

Methodology

The proposed system follows a structured pipeline:

1. **Data Collection**
   Collect student data such as attendance, marks, assignments, and participation

2. **Data Preprocessing**
   Handle missing values, remove outliers, and encode categorical data
3. **Feature Engineering**
   Generate useful features such as attendance percentage, average marks, and engagement score
4. **Model Training**
   Apply machine learning algorithms including:
   - Logistic Regression
   - Random Forest
   - Support Vector Machine (SVM)
5. **Model Evaluation**
   Evaluate models using accuracy and precision metrics
6. **Prediction & Visualization**
   Generate predictions and display results using graphs and dashboards

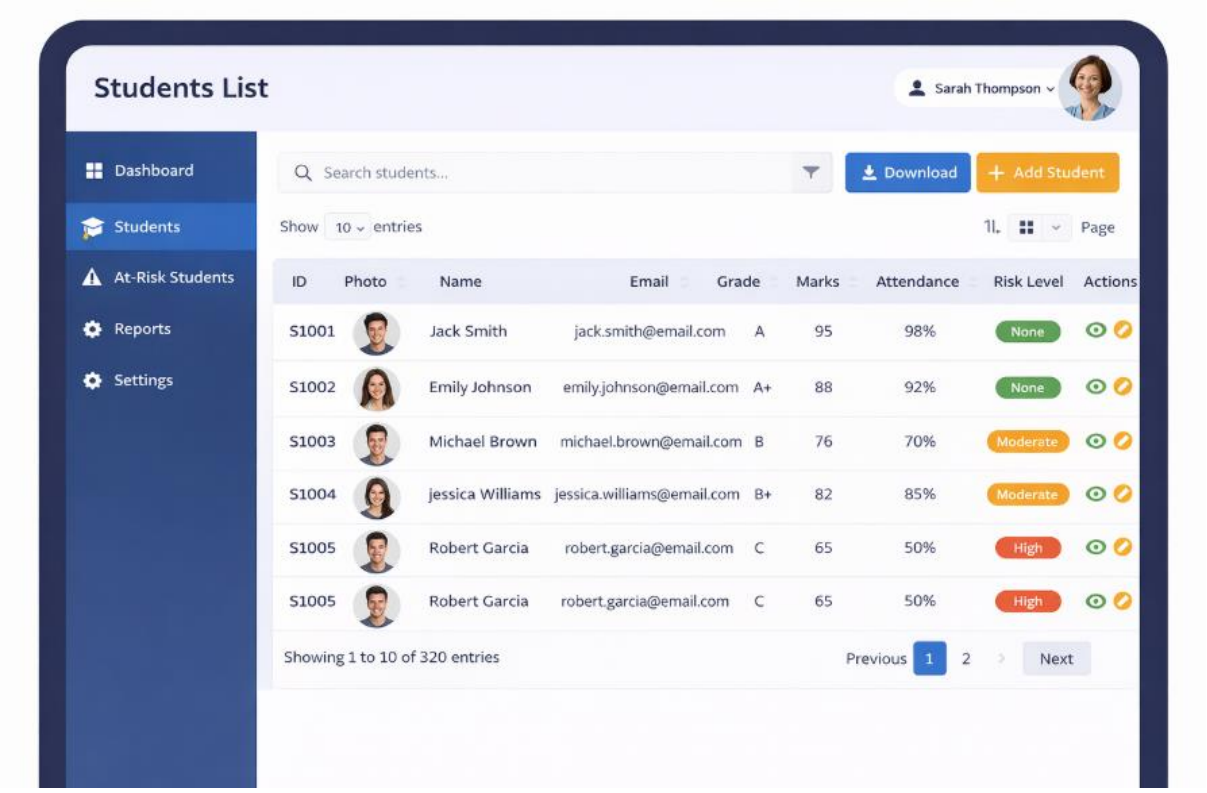


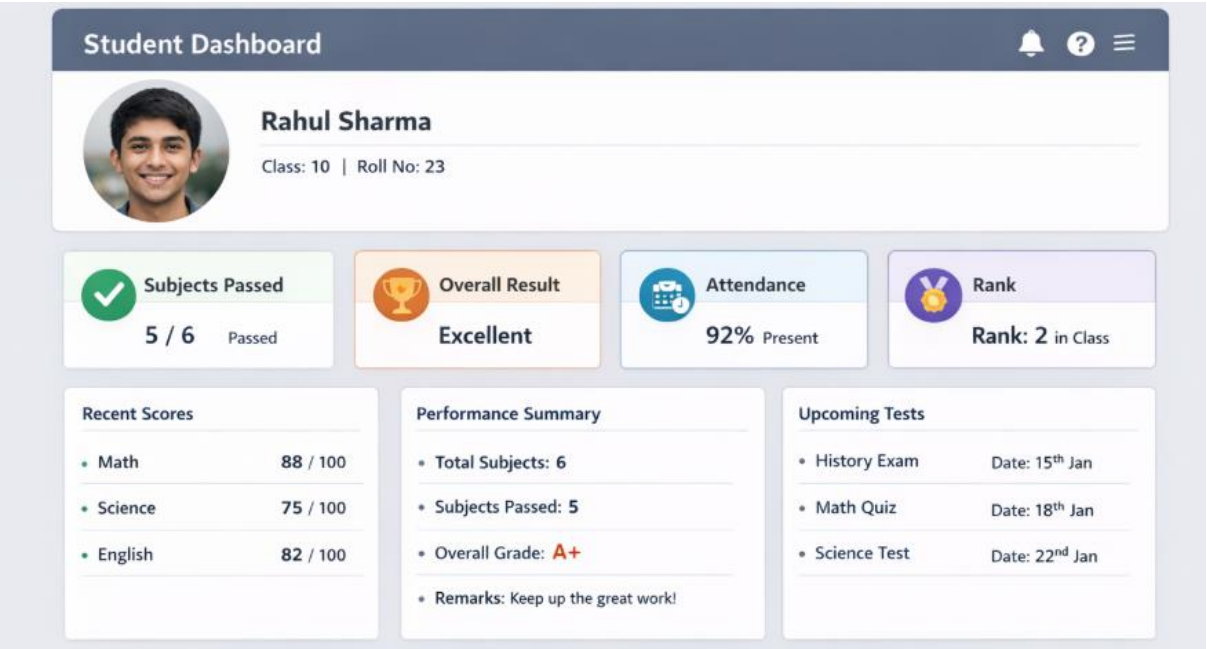
Tools & Technologies

- Programming Language: Python
- Libraries: Pandas, NumPy, Scikit-learn, Matplotlib
- Platform: Jupyter Notebook / Google Colab
- Interface: Streamlit (for web-based dashboard)

Expected Outcomes

- Accurate prediction of student academic performance
- Early identification of low-performing students
- Improved decision-making for educators
- Real-time performance monitoring
- Visualization of trends using graphs





(*these are sample pictures only*)

Applications

- Schools and colleges for performance monitoring
- Online learning platforms (LMS systems)
- Academic analytics and reporting systems
- Personalized learning systems

Conclusion

The proposed system demonstrates the effective use of machine learning techniques in the education sector. By analyzing student data and predicting performance in advance, the system helps identify at-risk students and enables timely intervention. This improves overall academic performance and supports data-driven educational practices.

Guide Signature: _____
Date: _____